

Name: Chinta Lakshmi Nivas Jaswanth

Module 1

1. Implementing the Singleton Pattern

Logger.java

```
public class Logger {  
    private static Logger instance;  
    private Logger() {  
        System.out.println("Logger instance created.");  
    }  
    public static Logger getInstance() {  
        if (instance == null) {  
            instance = new Logger();  
        }  
        return instance;  
    }  
    public void log(String message) {  
        System.out.println("LOG: " + message);  
    }  
}
```

SingletonTest.java

```
public class SingletonTest {  
    public static void main(String[] args)  
    {  
        Logger logger1 = Logger.getInstance();  
        Logger logger2 = Logger.getInstance();  
        logger1.log("First log message");  
        logger2.log("Second log message");  
        if (logger1 == logger2) {  
            System.out.println("Only one Logger instance exists.");  
        } else {  
            System.out.println("Multiple Logger instances exist.");  
        }  
    }  
}
```

2. Implementing the Factory Method Pattern

Document.java

```
public interface Document {  
    void open();  
}
```

WordDocument.java

```
public class WordDocument implements Document {  
    public void open() {  
        System.out.println("Opening Word Document");  
    }  
}
```

PdfDocument.java

```
public class PdfDocument implements Document {  
    public void open() {  
        System.out.println("Opening PDF Document");  
    }  
}
```

ExcelDocument.java

```
public class ExcelDocument implements Document {  
    public void open() {  
        System.out.println("Opening Excel Document");  
    }  
}
```

DocumentFactory.java

```
public abstract class DocumentFactory {  
    public abstract Document createDocument();  
}
```

WordFactory.java

```
public class WordFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new WordDocument();  
    }  
}
```

PdfFactory.java

```
public class PdfFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

ExcelFactory.java

```
public class ExcelFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

FactoryMethodTest.java

```
public class FactoryMethodTest {  
    public static void main(String[] args) {  
        DocumentFactory wordFactory = new WordFactory();  
        Document word = wordFactory.createDocument();  
        word.open();  
        DocumentFactory pdfFactory = new PdfFactory();  
        Document pdf = pdfFactory.createDocument();  
        pdf.open();  
        DocumentFactory excelFactory = new ExcelFactory();  
        Document excel = excelFactory.createDocument();  
        excel.open();  
    }  
}
```